Harmonic oscillator

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}kx^2 = \frac{\hat{p}^2}{2m} + \frac{1}{2}k\hat{x}^2$$

let $\omega = \sqrt{\frac{k}{m}}$, $\hat{p} = \tilde{p}\,(km)^{1/4}$, $\hat{x} = \frac{\tilde{x}}{(km)^{1/4}}$

$$\hat{H} = \sqrt{\frac{k}{m}}\,\frac{1}{2}\tilde{p}^2 + \frac{k}{2}\frac{\tilde{x}^2}{(km)^{1/2}} = \frac{1}{2}\omega\left(\tilde{p}^2 + \tilde{x}^2\right)$$

Note that $[\tilde{p}, \tilde{x}] = [\hat{p}, \hat{x}] = -i\hbar$

Introduce
$$\begin{cases} a = \dfrac{\tilde{x} + i\tilde{p}}{\sqrt{2\hbar}} \\ a^+ = \dfrac{\tilde{x} - i\tilde{p}}{\sqrt{2\hbar}} \end{cases} \Rightarrow \begin{cases} \tilde{x} = \sqrt{\dfrac{\hbar}{2}}\left(a + a^+\right) \\ \tilde{p} = -i\sqrt{\dfrac{\hbar}{2}}\left(a - a^+\right) \end{cases}$$

$$[a, a^+] = \frac{1}{2\hbar}\left[\tilde{x} + i\tilde{p}, \tilde{x} - i\tilde{p}\right] = \frac{1}{2\hbar}\left((-i)(i\hbar) + i(-i\hbar)\right)$$
$$= 1 \;!$$

$$\hat{H} = \frac{1}{2}\omega\frac{\hbar}{2}\left(-(a-a^+)^2 + (a+a^+)^2\right.$$

$$= \frac{1}{4}\hbar\omega\left(aa + aa^+ + a^+a + a^+a^+ - \right.$$
$$\left. - \left[aa - aa^+ - a^+a + a^+a^+\right]\right)$$

$$= \frac{1}{2}\hbar\omega\left(\underbrace{aa^+}_{1 + a^+a} + a^+a\right) = \hbar\omega\left(a^+a + \frac{1}{2}\right)$$

Next observe that

$$[\hat{H}, a^+] = \hbar\omega\left[a^+a, a^+\right] = \hbar\omega\left(a^+\underbrace{aa^+}_{1 + a^+a} - a^+a^+a\right)$$

$$= \hbar\omega a^+$$

$$[\hat{H}, a] = -\hbar\omega a$$

Therefore, if there is an eigenstate $|E\rangle$ such that $\hat{H}|E\rangle = E|E\rangle$

Than $\hat{H} a^+ |E\rangle = (\hat{H} a^+ - a^+ \hat{H} + \hat{a}^+ \hat{H})|E\rangle =$

$$= [H, a^+]|E\rangle + a^+ \hat{H}|E\rangle$$

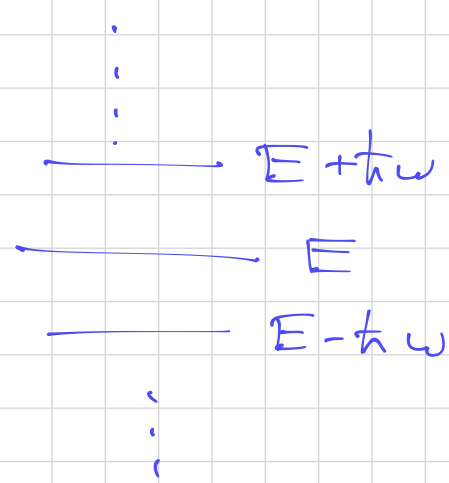
$$= \hbar\omega\, a^+ |E\rangle + a^+ E|E\rangle$$

$$= (E + \hbar\omega)\, a^+ |E\rangle$$

$\Rightarrow$ therefore state $a^+|E\rangle$ is an eigenstate, with energy $E + \hbar\omega$, too!

Similarly, $a|E\rangle$ is an eigenstate with $E - \hbar\omega$.

$\Rightarrow$ <u>ladder of states</u>



But the ladder can not continue down forever (why?!) $\Rightarrow$ there must be a state $|E_0\rangle$ such that $\hat{a}|E_0\rangle = 0$.

What is E_0?

$$\hat{H}|E_0\rangle = \hbar\omega\left(a^+ a + \tfrac{1}{2}\right)|E_0\rangle = \frac{\hbar\omega}{2}|E_0\rangle$$

$$\Rightarrow E_0 = \tfrac{1}{2}\hbar\omega.$$

Thus: we label eigenstates by their $\underbrace{a^+ a}_{\text{occupation number}}$ eigenvalue n.

$$\Rightarrow |E\rangle \to |n\rangle, \quad E_n \to \hbar\omega\left(n + \tfrac{1}{2}\right)$$

$$\hat{H}\,|n\rangle = \hbar\omega\left(n+\tfrac{1}{2}\right)|n\rangle$$

Note that $|n=0\rangle \sim |E_0\rangle$

$$\Rightarrow \quad |n\rangle \sim (\hat{a}^+)^n\,|0\rangle$$

Can normalize it as well, via

$$a^+|n\rangle = c_n\,|n+1\rangle$$

$$\Rightarrow \quad \langle n|\underbrace{a\,a^+}_{1+a^+a}|n\rangle = |c_n|^2 \quad \Rightarrow \quad c_n = \sqrt{n+1}$$

therefore $\hat{a}^+|n\rangle = \sqrt{n+1}\,|n+1\rangle$

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$$

(indeed, $\langle n|a^+a|n\rangle = |c_{n-1}|^2 = n$)

How does it look in space?

$$\hat{a}\,\psi_0 = 0 \quad \Rightarrow \quad (\hat{x} + i\hat{p})\,\psi_0 = 0$$

$$\left((km)^{1/4}\,\hat{x} + \frac{i}{(km)^{1/4}}\,\hat{p}\right)\psi_0 = 0$$

$$\left((km)^{1/2}\,\hat{x} + \hbar\frac{\partial}{\partial x}\right)\psi_0 = 0$$

$$k = \omega^2 m \qquad \hbar\frac{\partial}{\partial x}\psi_0 = -m\omega x\,\psi_0$$

$$\psi_0(x) = A\,e^{-\frac{m\omega}{2\hbar}x^2}$$

Fix A: $\quad A^2\int_{-\infty}^{\infty} dx\left(e^{-\frac{m\omega}{2\hbar}x^2}\right)^2 = 1$

$$A^2 \int dx\, e^{-\frac{m\omega}{\hbar}x^2} = 1$$

$$x = \sqrt{\frac{\hbar}{m\omega}}\, y \Rightarrow A^2 \sqrt{\frac{\hbar}{m\omega}} \overbrace{\int_{-\infty}^{\infty} dy\, e^{-y^2}}^{\sqrt{\pi}} = 1$$

$$\int dy\, e^{-y^2} \int dz\, e^{-z^2} = \int dy\, dz\, e^{-(y^2+z^2)} =$$

$$= 2\pi \int_0^\infty dr\, r e^{-r^2} = \pi \int_0^\infty d(r^2) e^{-r^2} = \pi$$

Thus $A = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$.

Eigenstates $\psi_n = \langle x | n \rangle$ are found by acting on ψ_0 with a^+.

Basic statistical mechanics

The quantum state of system is not known.

$$\langle O \rangle = \sum_i w_i \langle i | O | i \rangle$$

w_i: probability to be in $|i\rangle$

$|i\rangle$: orthonormal basis of $\hat{H}$

$$\sum_i w_i = 1$$

Microcanonical ensemble: the system is isolated, the energy is fixed $\rightarrow E$. all states with energy E are equally probable (ergodicity)

$\langle O \rangle = \mathrm{Tr}\{\rho O\}$ where the density matrix

$$\rho = \sum_i w_i |i\rangle\langle i| \quad \text{and} \quad \mathrm{Tr}(\rho) = 1$$

$$\rho = C\, \delta(\hat{H} - E)$$

Let's start thinking classically
(also helps to develop intuition)

isolated system of N particles in volume V, with total energy E.

E is held constant

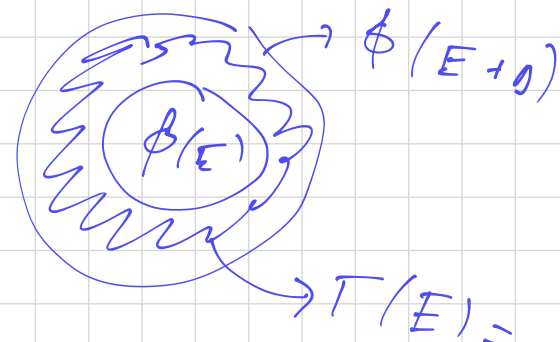
The distribution function $\rho(x,p) = \rho(H(x,p))$

$$\rho(x,p) = \begin{cases} \dfrac{1}{\Gamma(E,N,V)} & \text{for } E < H(x,p) < E+\Delta \\ 0 & \text{otherwise} \end{cases}$$

in the limit $\Delta \to 0$

The number of states (phase volume)

$$\Gamma = \underbrace{\int d^{3N}\vec{x} \int d^{3N}\vec{p}}_{E < H(\vec{x},\vec{p}) < E+\Delta}$$



For $\Delta \to 0$ $\quad \Gamma(E) = \dfrac{\partial \phi(E)}{\partial E}\Delta = \Omega(E)\cdot\Delta$

$\to$ density of states $\Omega(E) = \dfrac{\partial \phi}{\partial E} = \int d^{3N}\vec{x} \int d^{3N}\vec{p}\, \delta(E - H(x,p))$

and $\rho = \begin{cases} \dfrac{1}{\Omega(E)\Delta} \\ 0 \end{cases}$

<u>Boltzmann</u> entropy $S(E,V,N) = k_B \ln(\Gamma(E,V,N))$
$= k_B \ln(\Omega\cdot\Delta)$.

The object under log must be dimensionless

$\Rightarrow \quad S = k_B \ln\left(\dfrac{\Gamma}{\Gamma_0}\right)$

where $\Gamma_0 = h^{3N} N!$ $\to$ counting factor to account for indistinguishibility of the particles

$\hookrightarrow$ <u>Planck constant</u>

(without it we run into Gibbs paradox)

$S = k_B \ln(\text{Multiplicity})$

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N}, \quad \frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_{E,N}, \quad -\frac{\mu}{T} = \left(\frac{\partial S}{\partial N}\right)_{E,V}$$

$$dS = \underbrace{\frac{\partial S}{\partial E}}_{\frac{1}{T}} dE + \underbrace{\frac{\partial S}{\partial V}}_{\frac{P}{T}} dV + \underbrace{\frac{\partial S}{\partial N}}_{-\frac{\mu}{T}} dN$$

$$dE = T\,dS - P\,dV + \mu\,dN \qquad 1^{st} \text{ law of thermodynamics}$$

Free energy $F = E - TS$

$$dF = dE - d(TS) = dE - T\,dS - S\,dT$$

$$= -S\,dT - P\,dV$$

$$\Rightarrow \quad S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$$

$\Rightarrow$ let's try classical ideal gas:

$$H(\vec{x}_i, \vec{p}_i) = \sum_{i=1}^{N} \frac{\vec{p}_i^{\,2}}{2m} \qquad \text{in volume } V,$$

start with $\Phi(E) = \int\int \underbrace{d^{3N}\vec{x}}_{\prod_i d\vec{x}_i} \; d^{3N}\vec{p} = V^N \int d^{3N}\vec{p}$

spatial volume

$\sum \vec{p}_i^{\,2} \leq 2mE$

volume of 3N-dimensional sphere of radius $\sqrt{2mE} = R_E$

denote it as

$$V_{3N}(R_E) = R_E^{3N} \underbrace{\int_{\sum_i^{3N} y_i^2 \leq 1} dy_1 \ldots dy_{3N}}_{\equiv C_{3N}} = R_E^{3N} \cdot C_{3N}$$

momentum space volume

$$dy_1 dy_2 \ldots dy_{3N} = dV_{3N}(R) = C_{3N}\, 3N\, R^{N-1}\, dR$$

unknown coeff.

We find it by considering auxilary integral

$$J = \int_{-\infty}^{\infty} dy_1 \ldots \int_{-\infty}^{\infty} dy_{3N}\, e^{-(y_1^2 + y_2^2 + \ldots y_{3N}^2)} = (\sqrt{\pi})^{3N} = \pi^{3N/2}$$

$$J = \int_0^{\infty} 3N\, C_{3N}\, R^{3N-1}\, dR\, e^{-R^2} = \pi^{3N/2}$$

Let $R^2 = x \Rightarrow J = 3N\, C_{3N}$

$dx = 2R\,dR$

$dR = \frac{1}{2}\frac{dx}{R} = \frac{1}{2} x^{-1/2}\, dx$

$$J = \frac{1}{2}(3N)\, C_{3N} \underbrace{\int_0^{\infty} dx\, x^{-1/2} (x)^{\frac{3N-1}{2}} e^{-x}}_{\int_0^{\infty} dx\, x^{\frac{3N}{2}-1} e^{-x} = \Gamma\left(\frac{3N}{2}\right)} = \pi^{3N/2}$$

Γ-function $\Gamma(z) = \int_0^{\infty} dx\, x^{z-1} e^{-x}$

$\Rightarrow$ finally $C_{3N} = \dfrac{\pi^{\frac{3N}{2}}}{\frac{3N}{2}\Gamma\left(\frac{3N}{2}\right)}$

so that $\Phi(E) = V^N C_{3N} \left(\sqrt{2mE}\right)^{3N}$

$$\Omega(E) = \frac{\partial \Phi}{\partial E} = V^N C_{3N} (2mE)^{\frac{3N}{2}-1} \frac{3N}{2} \cdot 2m$$

Therefore the entropy is

$$S(E,V,N) \cong k_B \ln \left(\frac{\Phi(E)}{h^{3N} N!}\right)$$

there is not much difference between the volume and the surface area of 3N-dim. sphere for $N \gg 1$.

Stirling formula

$$\ln N! \approx N \ln N - N$$

$$S = k_B \ln\left(\frac{V^N (2mE)^{3N} C_{3N}}{h^{3N} N!}\right)$$

$$C_{3N} = \frac{\pi^{3N/2}}{\frac{3N}{2}\Gamma\left(\frac{3N}{2}\right)} = \frac{\pi^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)}$$

$$\ln C_{3N} = \frac{3N}{2}\ln\pi - \left(\frac{3N}{2}\ln\frac{3N}{2} - \frac{3N}{2}\right) =$$

$$= N\left\{\ln\left(\frac{2\pi}{3N}\right)^{3/2} + \frac{3}{2}\right\}$$

so that $S = k_B N \left\{\ln\left[\left(\frac{4\pi m E}{3h^2 N}\right)^{3/2} \frac{V}{N}\right]\right.$

Sackur-Tetrode equ.

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N} = k_B N \frac{3}{2} \cdot \frac{1}{E}$$

$$\Rightarrow E = \frac{3}{2} N k_B T$$

$$\frac{P}{V} = \left(\frac{\partial S}{\partial V}\right)_{E,T} = \frac{k_B N}{T} \Rightarrow PV = N k_B T$$